

Behind the Veil of Origin: Revisiting the Impacts of the French Headscarf Ban in Schools*

Sébastien Montpetit[†]

November 7, 2025

Abstract

This paper examines the impact of prohibiting the Islamic veil in schools on economic outcomes and long-run integration of Muslim women. Using a difference-in-differences design, I show that the 1994 directive instructing school principals to ban the veil led to a substantial decline in educational attainment among affected cohorts, with persistent consequences for employment and marriage market outcomes. An analysis of mechanisms suggests that these effects stem primarily from heightened perceptions of discrimination and mistrust toward the French school system, rather than shifts in parental educational investments. I also show that misclassification of religion in prior work introduces substantial bias. Despite the adverse economic consequences, the affected cohorts exhibit stronger identification with France but also higher levels of religiosity, suggesting a mixed long-run impact on cultural assimilation.

Keywords: headscarf ban, religious identity, women's education, cultural integration, marriage market, misclassification bias

JEL Codes: I28, J16, J15, Z12

*I thank Philippe Bontems, Jean-Paul Carvalho, Marie Connolly, Amira Elasra, James Fenske, Antoine Jacquet, Jean-William Laliberté, Marie-Louise Leroux, Lea Nassal, François Poinas, and Paul Seabright for useful suggestions and comments on this project. I am grateful for financial support from TSE-UMR and the Institute for Replication. All remaining errors are my own.

[†]Department of Economics, University of Warwick (email: sebastien.montpetit@warwick.ac.uk).

1 Introduction

Since the mid-20th century, the share of Muslims in the European population has been steadily rising, and their religious practices have increasingly come into tension with prevailing Western norms. In particular, the Islamic veil is often perceived both as a symbol of women’s subordination and as a challenge to state secularism. In response, roughly one-third of European countries have adopted policies restricting the wearing of Islamic clothing in public spaces ([Abdelgadir and Fouka 2020](#)). While such regulations are often justified as preserving the majority culture, their effects on the targeted populations are theoretically ambiguous. On the one hand, if women do not choose to veil willingly, bans may liberate them from constraining social norms. On the other hand, if veiling provides significant religious benefits, prohibitions may lead to more social exclusion and segregation ([Carvalho 2013](#), [Jacquet and Montpetit 2024](#)).

To address this question, I study the case of France, the only developed country to prohibit the Islamic headscarf in public schools for both pupils and employees.¹ In 1994, Minister of Education François Bayrou issued a circular (henceforth, the Bayrou circular) instructing school principals to ban the veil. The policy was gradually adopted by schools over the following years and was ultimately codified into law in 2004.

I extend the current evidence on veil bans by analyzing the effects of these two policy shocks on both economic and social integration. A key advantage of my approach is the ability to directly observe individuals’ religion using unique survey data, rather than relying on origin as a proxy. To identify the impact of these reforms, I implement a difference-in-differences design comparing cohorts of Muslim women who reached the mandatory schooling age (16 years old) before and after the policy.

I document a large negative impact of the circular issuance on Muslim women’s likelihood of completing high school. The point estimates suggest a decline in the high-school completion rate above 18% of the pre-treatment mean and is robust to a range of sensitivity checks. This large negative effect dissipates after six cohorts have reached the compulsory-schooling age under the ban. I further show—using detailed data on respondents’ educational trajectories—that this impact translates into reduced total years of schooling.

I next investigate the mechanisms through which the headscarf ban reduced schooling. Consistent with qualitative evidence in [Abdelgadir and Fouka \(2020\)](#), Muslim women

¹Very few other countries implemented headscarf bans for students. Turkey banned the wearing of the veil on university campuses in 1997, but the regulation was repealed by the pro-Islamic conservative Justice and Development Party (AKP) in 2010 ([Corekcioglu 2021](#)). To my knowledge, Kazakhstan (since 2023) is the only other country to ban headscarves for both teachers and pupils. Other European countries such as the Netherlands, Belgium, Denmark, Bulgaria, Norway, and Switzerland prohibit the wearing of face-covering garments (i.e. the niqab and burqa), but not the headscarf (i.e. the hijab). Recently, the provincial government of Quebec—the second-most populous Canadian province—proposed Bill 94 which would ban the veil for pupils.

exposed to the directive were 13 percentage points (38%) more likely to report religious discrimination. The alleged discrimination is driven by perceptions of unfair treatment at school, which increases by 12.3 percentage points (35%). The Bayrou circular is also associated with an increase in mistrust of the French school system among affected Muslims, suggesting that the disruption in girls' educational trajectories reflects a hostile school environment.

An alternative explanation is that the ban prompted strategic reactions from identity-concerned Muslim parents who perceived the policy as a threat to their religious traditions (e.g. [Dahl et al. 2022](#)). However, I find little evidence for such parental responses. First, the effects are similar across families which devoted more effort in transmitting their religion, indicating that the impact is not concentrated among the most pious. Second, I show that Muslim parents did not reduce their educational investments in their daughters. Affected girls were no less likely to receive homework help from family members, nor more likely to attend private schools where the ban did not apply. Third, heterogeneity analyses show that the decline in schooling is more pronounced among girls from more traditional families in an economic sense (male-breadwinner households), rather than those in families making greater efforts to transmit religion.

Finally, I examine the long-run effects of the ministerial circular on religiosity, social integration, and economic outcomes. The decline in schooling leads to higher dropout rates and a substantial 10.5 percentage point reduction in employment later in life (16% of the mean) along with spillovers into the marriage market. Consistent with extensive evidence on assortative mating ([Choo and Siow 2006](#), [Chiappori et al. 2017](#)), affected Muslim women are more likely to be married to an unemployed partner. As expected, given their earlier school exit, they are also more likely to have children.

Turning to cultural and social integration, I find no clear evidence of a religious backlash. In contrast with findings from Turkey ([Sakalli 2019](#)) and Indonesia ([Bazzi et al. 2025](#)), I detect no increase in religiosity in adulthood. However, I document a rise in veiling rates in public spaces, which could indicate a symbolic response to the prohibition. I also find that affected cohorts are more likely to form most of their friendships within their own religious group. Despite these indications of increased segregation, these cohorts also express stronger identification with French society. Taken together, these results suggest a simultaneous strengthening of both national and religious identities ([Beaman 2016](#), [Abdelgadir and Fouka 2020](#)).

I build on two earlier studies of the French headscarf prohibition that relied on origin proxies ([Abdelgadir and Fouka 2020](#), [Maurin and Navarrete-Hernández 2023](#)). I find that accurately identifying the targeted group in this setting reveals several new insights into the effects of religious prohibitions in secular contexts. Using a different data source than in those studies, I confirm a positive effect of the Bayrou circular on high-school graduation rates for individuals of African origin. Yet for the actual treatment group—Muslim

women—the effect is of opposite sign. I show that these patterns also hold for the number of years of schooling, a new result in the literature on the French headscarf ban.

The stark contrast between estimates based on religious affiliation and those relying on origin highlights substantial bias from treatment misclassification in earlier studies. I show that a wrongly signed estimate is consistent with recent findings in econometrics of difference-in-differences with a misclassified treatment (Denteh and Kédagni 2022, Negi and Negi 2025). Moreover, I find no clear effect of the 2004 law, consistent with the prohibition having already been widely implemented following the Bayrou circular.

This study first contributes to the literature on assimilation and integration policies. I show that the French headscarf ban reduced the schooling outcomes of Muslim women, with evidence pointing to increased discrimination as the main channel. These findings echo previous work on other forms of *assimilationist* regulation, which often backfire (e.g. Lleras-Muney and Shertzer 2015, Fouka 2020). In the context of veil bans, Corekcioglu (2021) and Lu and Yurdakul (2025) document positive effects of lifting prohibitions in the Turkish public sector on female employment. By contrast, *integration* policies—such as easier access to citizenship—are typically associated with stronger labor market attachment and greater social integration of immigrants (see Gathmann and Garbers (2023) and Fouka (2022; 2024) for recent reviews, as well as Kamel (2025) on Arab immigrants following the *Dow v. United States* ruling). A notable exception is Dahl et al. (2022), who show that the introduction of automatic birthright citizenship in Germany lowered life satisfaction, self-esteem, and social integration among Muslim girls. Consistent with the broader evidence on assimilationist policies, my results suggest that headscarf bans are unlikely to promote economic integration either.

Second, this paper contributes to the literature on the interplay between education and identity. I show that restricting the ability of minorities to express their identity in schools can generate unintended consequences. Rather than fostering Muslim women’s integration into secular society, the French headscarf ban reduced their educational attainment. This finding is consistent with economic theory suggesting that veil bans may hinder the economic integration of pious women (Carvalho 2013), as well as with recent models showing that marginalized cultural communities may underinvest in human capital when the secular content of mainstream education is strengthened (Carvalho et al. 2017, Bisin et al. 2023, Carvalho et al. 2024). Related empirical work demonstrates that changes in school curricula can shape identity and attitudes (Cantoni et al. 2017, Squicciarini 2020, Bazzi et al. 2025). Closer to my setting, Sakalli (2019) shows that the secularization of Turkish schools lowered educational attainment and increased religiosity, particularly in pious districts prior to the reform.² My study differs in that I examine a minority context, where it was not the content of education but rather the conditions of schooling that were

²Similarly, Benzer (2022) finds that the subsequent reintroduction of Islamic schools in Turkey, where the headscarf is permitted, improved girls’ educational attainment.

altered.

Third, this paper connects to the growing literature on reproducibility and replicability in economics. A key finding in this literature is that many published results fail to replicate, raising questions about the reliability of causal claims in empirical research (Gertler et al. 2018, Huntington-Klein et al. 2021; 2025, Brodeur et al. 2024). Related evidence of p-hacking and the lack of transparency in economics research (e.g. Brodeur et al. 2016; 2020) has fostered new research norms that treat replication as a crucial diagnostic tool (Miguel 2021). I contribute to this literature by reassessing the impacts of a reform with significant implications for the integration of religious minorities in Western countries. Specifically, I perform a *direct replication* of two prior studies on the French headscarf ban using an alternative data source that allows for correct classification of the treatment group, while following similar empirical procedures.³ I show that misclassifying the targeted group can substantially alter conclusions about the policy’s effects on economic integration. While my results align with Maurin and Navarrete-Hernández (2023) when using origin as a proxy for Muslim affiliation and their empirical specification, the effects on the actual treatment group are of opposite sign.

The rest of the article is structured as follows. Section 2 describes the institutional context. Section 3 presents the data sources along with summary statistics. Section 4 evaluates the impact of the ban on Muslim women’s educational attainment, presents my analysis of the underlying mechanisms, and discusses misclassification biases in previous papers. Section 5 analyzes impacts on long-term outcomes. Finally, Section 6 concludes.

2 Institutional context

2.1 The French headscarf ban in schools

The wearing of the Islamic veil has been a burning issue in France since at least three decades. In 1989, the “*affaire des foulards*” (headscarf affair) garnered nationwide attention when three girls were expelled from their middle school for refusing to remove their headscarves. The incident sparked heated debates and was followed by similar disputes in other schools. Eventually, the affair was settled after the highest French administrative court (the *Conseil d’État*) ruled in favor of the expelled girls (Scott 2009). In its ruling, the Council stated that banning the wearing of signs of religious affiliation by students in public schools was against their freedom of religion.

The 1994 Bayrou circular. Five years later, following the election of a right-wing government, the minister of education, François Bayrou issued a circular asking school

³Different types of replications and their definitions are discussed in Dreber and Johannesson (2025).

principals to prohibit conspicuous religious symbols worn by students.⁴ To justify the government's position, the document insists on the distinction between conspicuous symbols and discreet signs. The Minister argued that conspicuous symbols are "in themselves acts of proselytizing" and should thus be prohibited in public schools. This interpretation of the Council's ruling contrasts with its original meaning, which stated that as long as the student's behavior was not disrupting class activities, one should not be refused admission to school for wearing a veil. Despite some opposition from the *Conseil d'État*, most school principals decided to follow Bayrou's recommendation and to adopt the ban over the following years.

To help implement the bans, Simone Veil, the minister of social affairs, appointed a woman of North-African origin, Hanina Chérifi, as mediator to handle problems on the ground. In the school year that followed the circular (issued in September 1994), around 3,000 cases required an intervention from the mediator with only 139 leading to exclusions. The number of interventions quickly dropped to 1,000 in 1996 and to about 150 in 2002, suggesting that most establishments implemented the ban soon after the circular was issued.

Despite these small numbers, all the media and political attention likely changed the schooling environment for Muslim pupils beyond the management of these cases. There was, "during this period, renewed significant media coverage of the hijab issue, with [...] a shift in tone in the press commentary. [...] And once again, few women, apart from the occasional headscarved girl, were asked what they thought" (Winter 2009, p. 185-187). Such a one-sided debate might have created a hostile schooling environment against Muslim pupils. The potential impacts on the school environment are explored in Section 4.4.

The 2004 law. Partly fueled by concerns about terrorism, President Jacques Chirac appointed a commission in July 2003 to evaluate the possibility of enshrining the 1994 circular into law. Once again, the public debate was fierce. On the one hand, proponents of the ban argued that the policy could not only "free" Muslim girls from religious pressures, but also that headscarves "infringed on the liberty of conscience of other pupils" (Abdelgadir and Fouka 2020, p. 4). On the other hand, critics replied that it would rather impede the integration of Muslim girls by excluding them from public education.

The commission ultimately recommended a ban on conspicuous religious signs in public schools. The French government voted the ban in March 2004, which was enforced in October 2004. As for the ministry circular, the Islamic veil was the main target of the law and Muslim pupils were the main group affected in practice. A report from Chérifi (2004) based on fieldwork in four school academies shows that, in 2004-2005, only 639

⁴In France, a ministry circular is a governmental document which gives a clarification or an interpretation of the law or establishes guidelines for civil servants.

students showed up to school wearing a conspicuous religious sign, less than half than in the previous academic year. While most cases were resolved through dialogue, about 200 of these 639 students switched to private schools or opted for distance learning (Mattei and Aguilar 2016).

2.2 The French educational system

The headscarf ban targets pupils at the primary and secondary levels, but does not apply to students attending college. In France, pupils enter elementary school at age 6 and it lasts for 5 years (until age 11). Then, they attend middle school (*collège*) for four years (until age 15) and school attendance is mandatory until age 16. Therefore, potential impacts of the ban should manifest after middle school. After middle school, students enter high school, either to pursue a vocational degree (CAP/BEP) in a professional high school (typically for two years) or to prepare for the *baccalauréat* in a general or a technological high school, which lasts three years). The *baccalauréat* is the diploma required for continuing in higher education. Following previous studies of the French headscarf ban, completion of this degree is the primary outcome of interest.

3 Data and descriptive analysis

3.1 Data sources and sample

My primary data source is the two waves of the *Trajectoires et Origines* surveys (henceforth TeO; Beauchemin et al. (2016; 2023)). Conducted in 2008–2009 and 2019–2020 by the French National Institutes for Demographic Studies, the TeO surveys targeted adults between 18 and 60 years old residing in metropolitan France. Purposefully oversampling immigrants and minorities, it includes 3,033 and 3,519 women who identify as Muslim in the first and second waves respectively. To the best of my knowledge, these are the largest samples of this kind in France.⁵ When including Muslim men and other religious groups, the entire surveys contain more than 21,000 observations each.

The TeO datasets are a comprehensive source of information on various aspects of respondents’ lives, including living conditions (such as employment, education, housing, commune of residence, and health), social life (such as migration history, language use, family, and children), and public life (such as political views, experiences of discrimination, and social relationships). A unique feature for a French survey of this scope is a section on religion, since the collection of individual religious information is tightly regulated

⁵Two surveys conducted by private firms, namely Institut Montaigne (2016) and Institut Français d’Opinion Publique [IFOP] (2019), have much smaller sample sizes—slightly above 1,000 individuals of Muslim origin, both genders included. Moreover, the TeO surveys are much richer on many other dimensions and response rates are higher given its mandatory nature.

in France. This section includes information such as religious affiliation, measures of religiosity, religious symbols worn, and intergenerational religious transmission. This is a key advantage over the French Labour Force Surveys (LFS), the data source used in previous studies ([Abdelgadir and Fouka 2020](#), [Maurin and Navarrete-Hernández 2023](#)). Indeed, the LFS only offers a proxy for religious affiliation, namely the parents' place of (and nationality at) birth. I also observe origin in the TeO surveys, which I use to replicate the previous papers.

Measurement of religion. A key contribution of this paper is to properly identify the treatment and control groups. The empirical analysis relies on an actual measure of religious affiliation along with measures of religiosity.⁶ Response rates to the religion section of the survey are very high: religious affiliation is observed for more than 98% of the sample in each wave. Other sections of the survey were mandatory.

First, for religious affiliation, respondents are asked whether they currently have a religion. If they answer “yes”, then they are asked which one and the interviewer notes the exact answer given. Particular denominations within each religious family are then pooled into 12 groups in the survey data. For the main empirical analysis, I further pool religious denominations into five groups to use as controls: Atheists (no religion), Catholics, Other Christians, Muslims, and Others.⁷ The same questions are asked about other family members, namely the father, the mother, and the partner.

Second, the TeO surveys include a set of questions about individual religiosity. My preferred measure is the frequency of attendance of religious ceremonies, a standard measure of religiosity which focuses on religious practice ([Iyer 2016](#)). I also consider other measures of individual religiosity: the self-reported importance of religion in the respondent's life and whether she uses her religion to self-identify. To exploit the common variation in these measures, I also build a latent index of individual religiosity as in [Jacquet and Montpetit \(2024\)](#). Details are provided in Appendix [A.1](#).

Third, for veiling behavior, I use the following question from the TeO survey:

In your daily life, do you wear in public a piece of clothing or jewelry that might evoke your religion? (1) Never (2) Sometimes (3) Always

If applicable, respondents were subsequently asked to report which religious symbols they wear. Answers were later sorted by the survey institute into four categories: jewelry, clothing, headcoverings, or others. Because they visibly signal religion and are the ones

⁶I do not use veiled women as the treatment group because veiling behavior is measured at the time of the survey and thus much later in life. Evidence in Indonesia suggests that improved economic opportunities induced women to veil ([Shofia 2022](#)). Therefore, not only is veiling not a permanent exogenous trait, but is potentially an outcome in my setting.

⁷The “Others” category is mostly composed of Buddhists, Jews, and Hindu/Sikh.

usually targeted by secular policies, I group the clothing and headcoverings categories together as *conspicuous symbols*.⁸

Educational attainment. My main outcome variable is schooling, which I measure in two ways. The TeO surveys report the specific grade level at which the individual left school up to the Master’s (*BAC+5*) level. I use this information to reconstruct both high-school completion and the number of years of schooling (from elementary school), capped at twenty years.⁹

Sample selection. The main empirical analysis focuses on the effects of the 1994 circular on girls reaching puberty (15 years old) around its issuance. Because individuals aged less than 20 years old might still be in high school, I focus on women aged at least 20 years old. This leads me to restrict the sample to individuals born between 1971 and 1987 so as to have similar numbers of observations on both sides of the cohort threshold. Given these restrictions, I end up with a working sample of 7,758 women—and 7,261 men for placebo exercises.

3.2 Descriptive analysis

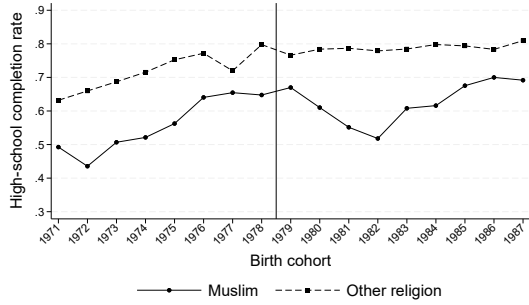
I start the empirical analysis by comparing the characteristics of Muslim women to those of other French-born women. Table A1 contains these statistics. Muslim women display less favorable economic outcomes than their non-Muslim counterparts. In terms of educational attainment, the main outcome of interest, Muslim women are twice as likely to have no education and are also substantially less likely to pursue in higher education. Only one third of Muslim women have completed a college degree while more than half of non-Muslim women graduated in higher education. This translates into a difference of one year of schooling. Muslim women also substantially differ from other women on various aspects of their lives such as religious characteristics. For example, they display higher levels of religiosity and are more likely to report that their parents invested in their religious education.

To understand whether the headscarf ban might have played any role in Muslim women’s education, I first plot the time series of my main outcomes of interest in Figure 1, separately for Muslim women and women of other religious affiliation. A striking pattern emerges: while the educational attainment of Muslim women born in the 1970s was somewhat catching-up over that of women of other religious groups, there is a significant break in this trend from the 1980 cohort. The drop in high-school completion rates (panel A) is substantial and it increases over the following three affected cohorts. Thus, at first glance, this suggests that the issuance of the circular abruptly delayed the ongoing

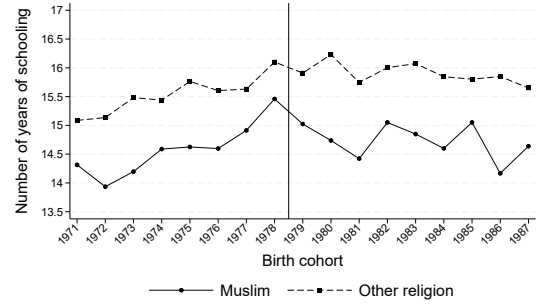
⁸I use the terms “veiling” and “conspicuous symbols” interchangeably in the remainder of the paper.

⁹Only about 11% of the sample has completed at least a Master’s degree.

Figure 1: Evolution of educational attainment of Muslim and non-Muslim women across birth cohorts, 1971-1987



(a) High-school completion



(b) Years of schooling

Note: Evolution of the high-school completion rate across birth cohorts of Muslim and non-Muslim women born in France. On the right of the vertical line are cohorts who reach puberty after the 1994 Bayrou circular asking school principals to prohibit the headscarf in schools.

reduction in educational gaps between the two groups. This negative trend appears to be short-lived as it then bounces back to that observed in the cohorts of the 1970s. For years of schooling (panel B), a similar decrease occurs, and it coincides with the first cohort reaching the mandatory-schooling age after the circular issuance. However, in contrast with the trend in the high-school graduation rate, the drop somewhat persists in the medium run. To move towards a causal interpretation, I estimate difference-in-differences models in the next section.

4 Impact on educational attainment

This Section covers the main empirical results. After describing the empirical strategy in Section 4.1, I present the main results in Section 4.2. Then I perform robustness checks (Section 4.3), investigate potential mechanisms (Section 4.4), and assess heterogeneity in policy impacts (Section 4.5). In Section 4.6 I compare my results with previous studies of the French headscarf ban and provide a detailed discussion of misclassification bias.

4.1 Empirical strategy

The main empirical specification is a standard difference-in-differences design, which compares cohorts of women of different religions reaching the age at which school is no longer mandatory before or after the 1994 circular was issued.

My approach differs from those of [Abdelgadir and Fouka \(2020\)](#) and [Maurin and Navarrete-Hernández \(2023\)](#) (henceforth AF and MN) in one crucial dimension, which is that I observe respondents' religious affiliation. I can thus identify Muslim women. In contrast, the empirical strategies used in those papers, summarized in Table 1, use proxies of religious affiliation, namely the father's country of birth or the father's nationality at

birth. The age threshold chosen to define treated cohorts (third column) and the specific policy being studied (fourth column) also differ between the two studies.

In this paper, as in MN, I focus on the 1994 circular because most schools already implemented bans in the first years after 1994 (Chérifi 2004). As for the cohort threshold defining treatment, I do not follow previous papers. While MN uses 15 years old as the age threshold—arguing that this captures puberty, and hence the age of veiling—, there is no evidence that girls decide whether to veil at that age. Moreover, puberty usually occurs earlier: De La Rochebrochard (1999) documents that the median age of puberty in France among the 1975-1978 cohorts is 13.1 years old. AF use the usual age of ending high-school as the age threshold. This choice is also rather arbitrary because the ban also applies in middle-school. A more natural cutoff in this setting is the age at which school is no longer mandatory and dropping-out becomes possible, namely 16 years old. I therefore consider individuals born 1979 or later as being in the treated cohorts.

Formally, for individual i in birth cohort c , I estimate the following model:

$$Y_{i,c} = \alpha + \beta_1 \text{Muslim}_i \times \text{Post}_c + \beta_2 \text{Muslim}_i + \gamma_c + X'_{i,c} \delta + \varepsilon_{i,c} \quad (1)$$

where Post_c is an indicator variable taking the value of 1 if the individual is born after 1978 (i.e. reaches age 16 after 1994), Muslim_i indicates Muslim affiliation, and γ_c is a full set of cohort dummies. In my simplest specification, the vector of controls X contains a dummy for the second survey wave and dummies for the other religious groups. I also control for a Muslim-specific linear trend to account for the reduction of schooling gaps observed in pre-1979 cohorts in Figure 1.

In another specification, I further control for the respondents' living conditions when aged 15 years old. That is, I include indicators for whether the individual's father was working, the mother was working, and dummies for the *département* of residence. The main outcome variables Y are a dichotomous variable taking the value of one if individual i has completed high school (a *baccalauréat*) or the number of years of schooling.

Table 1: Empirical strategies used in previous studies of the French headscarf ban

Study	Policy	Age threshold	Data	Muslim proxy	Clustering level
Abdelgadir and Fouka (2020)	2004 law	19 y.o. (end of <i>baccalauréat</i>)	LFS 05-12	Father's birthplace	Father's birthplace
Maurin and Navarrete H (2023)	1994 circular	15 y.o. ("puberty")	LFS 05-19	Father's nationality	Father's nationality $\times$ <i>département</i> of birth

Note: This table presents the empirical strategies used in previous studies of the impact of the French headscarf ban on the high-school graduation probability of Muslim women. The acronym LFS refers to the French Labour Force Surveys.

Inference. The appropriate method for calculating standard errors is yet another source of disagreement between the two previous studies. While AF cluster standard errors at the father’s birthplace level, MN do not use this approach, arguing that it yields a small number of clusters. However, their clustering level may not be appropriate either since they do not cluster at the level at which treatment varies (Abadie et al. 2023). Here, treatment varies at the religion (or father’s origin) level, but not over the individual’s birthplace within France.¹⁰ Therefore, I rather cluster standard errors at the religion (or origin) level as in AF. Since this leaves me with few clusters, I also report p -values calculated using the wild subcluster bootstrap procedure of MacKinnon and Webb (2018) accounting for the small number of clusters in difference-in-differences designs.¹¹

4.2 Main results

As shown in Section 3.2, the time series of educational attainment suggests that the 1994 circular induced a strong decline in schooling among Muslim women. To confirm the graphical evidence, I estimate the difference-in-differences model (1). Results are reported in columns (1) and (2) of Table 2.

In Panel A, I first focus on the effect of the circular on the high-school completion rate, the main outcome studied in previous papers. Consistent with the observed trends, I find that the ministerial circular causes an average decline in the probability of Muslim women to complete high school of 11 percentage points. This effect is economically large in magnitude as it represents 18% of the pre-ban mean for this group. The point estimate remains similar when controlling for the woman’s living conditions when she was 15 years old (column 2).

To further validate the impact on educational attainment, in Panel B, I exploit the richness of the TeO data to evaluate the impact on years of schooling. Precisely measuring years of completed education is not possible with the Labour Force Surveys and thus these results are novel in the literature on this reform. In line with the negative effect on high-school completion, the issuance of the circular is associated with a drop in years of schooling of more than 0.4 year (3% of the mean). In Appendix Table A10, I find that there is no discernible impact on completion of vocational high school (CAP/BEP), suggesting that these effects are driven by increased drop-out.

In columns (3) to (6) of Table 2, I assess how my results compare with previous studies. To do so, I estimate the impact of the circular using proxies for religious affiliation used

¹⁰Maurin and Navarrete-Hernández (2023) justify their clustering level by saying that it accounts “for potential correlation of residuals within groups of individuals with a similar background” (p. 84). Such clustering strategy is at odds with standard practice in economics (Abadie et al. 2023).

¹¹In difference-in-differences designs with few treated clusters, MacKinnon and Webb (2017) show that the wild cluster bootstrap procedure of Cameron et al. (2008) used in Abdelgadir and Fouka (2020) for robustness under-rejects the null hypothesis of null effects. The subcluster bootstrap procedure of MacKinnon and Webb (2018) is preferred in this setting.

Table 2: Impact of the 1994 ministerial circular on educational attainment

Measure of treatment:	Proxies used in previous studies [†]					
	Religious affiliation		Father's nationality (MN)		Father's birthplace (AF)	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: High school completion</i>						
Muslim _i × Post _c	-0.111*** (0.011) [0.026]	-0.102*** (0.018) [0.122]	0.016 (0.017) [0.534]	0.022 (0.018) [0.389]	0.022 (0.026) [0.523]	0.030 (0.024) [0.343]
Muslim _i	-0.104*** (0.031) [0.098]	-0.084*** (0.026) [0.107]	-0.165*** (0.038) [0.145]	-0.146*** (0.037) [0.159]	0.012 (0.031) [0.816]	0.010 (0.028) [0.812]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.609	0.609	0.622	0.622	0.637	0.637
N	6,623	6,623	6,544	6,544	6,596	6,596
R ²	0.072	0.103	0.085	0.112	0.084	0.111
<i>Panel B: Years of schooling</i>						
Muslim _i × Post _c	-0.446* (0.236) [0.527]	-0.458* (0.241) [0.515]	0.139* (0.072) [0.212]	0.169** (0.065) [0.103]	0.047 (0.113) [0.751]	0.103 (0.101) [0.431]
Muslim _i	-0.215 (0.310) [0.678]	-0.099 (0.247) [0.816]	-1.363*** (0.245) [0.069]	-1.250*** (0.254) [0.085]	0.240 (0.180) [0.388]	0.192 (0.181) [0.488]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	14.952	14.952	14.966	14.966	15.147	15.147
N	6,416	6,416	6,341	6,341	6,390	6,390
R ²	0.064	0.095	0.079	0.104	0.078	0.105

Note: This Table reports regression estimates of the impact of the Bayrou circular on educational attainment of Muslim girls using various measures of the treated group. Means of the dependent variable in the treatment group over pre-1979 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In even columns, additional control variables are indicators for whether the individual's father was working, the mother was working, and dummies for the *département* of residence at age 15. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[†] MN refers to [Maurin and Navarrete-Hernández \(2023\)](#) and AF refers to [Abdelgadir and Fouka \(2020\)](#).

in previous papers (see Table 1). I find that, when using origin proxies, the effects on schooling are positive and small in magnitude.

In Appendix Table A2, I further show that I can replicate the main result of [Maurin and Navarrete-Hernández \(2023\)](#) in the TeO data when using their 15-year old age threshold. They find that the issuance of the circular had a positive impact on girls of African origin, specifically of women whose father's has an African nationality. I obtain an estimate that is lower in magnitude, but similar to that found in that paper (4.7 versus

7.8 percentage points).

Results in columns (5) and (6), however, contrast with [Abdelgadir and Fouka \(2020\)](#), who find no effect of the Bayrou circular. As shown in Table 1, their study defines treatment using an alternative age threshold, namely adulthood. I interpret the absence of an effect among cohorts reaching age 19 before and after the circular as reflecting that any impact would likely have occurred earlier. It is also likely that these women had already made their veiling decisions, since according to Islamic precepts, girls are expected to begin veiling at puberty.

One important point to note, however, is that the statistical significance of the results is quite sensitive to the method used to calculate standard errors. When accounting for the small number of clusters using the wild subcluster bootstrap, some estimates—especially for years of schooling—have high p -values and thus lose statistical significance. However, this correction of p -values was not made in the two previous papers (see Table 1). My results using the standard clustering approach are thus more comparable to the previous evidence.

Threats to identification. The main identification assumption for a causal interpretation is the standard parallel-trends assumption in that Muslim and non-Muslim women’s educational attainment would have evolved the same absent the circular. To support the parallel-trends assumption, I estimate an event-study model, which takes the form:

$$Y_{i,c} = \alpha + \sum_{\substack{c=1971 \\ c \neq 1979}}^{1987} \beta_c \text{Muslim}_i \times D_c + \gamma_r \text{Muslim}_i + \gamma_c + X'_{i,c} \delta + \varepsilon_{i,c} \quad (2)$$

where D_c is an indicator taking the value of 1 if the individual is in birth cohort c . I use the same set of controls as in the difference-in-differences analysis.

Results are reported in Figure 2. For high-school completion, except one aberrant cohort (1975 for high-school completion and 1976 for years of schooling), the treatment and control groups appear to evolve similarly in the pre-treatment periods. The two groups evolve similarly from 1976 up to the first treated cohort. The graph also reveals that the average impact in Table 2 masks important dynamics. Consistent with the time series (Figure 1), the initially large negative impact vanishes from the 1985 birth cohort. I observe similar dynamics for the number of years of schooling, further validating my findings.

Another threat to identification arises if there is another shock differentially affecting Muslim women born in 1980. This is unlikely for two reasons. For one, the main other episode that spurred discrimination against Muslims in this period is the September 2001 attacks in New York City, which occurred several years after the circular. Second, the other change in the educational system for the 1980 cohort was a reform of vocational

high school. This policy reduced the number of years required to complete the vocational degree by one year for some tracks in 2008 and then for all occupational tracks in 2009. It also introduced catch-up exams for vocational high school students in their final year. Although ethnic (and religious) minorities tend to be overrepresented in vocational tracks (Belzil and Poinas 2010), there is no particular reason to believe that this policy change would differentially affect male and female students. Among other robustness checks, I use males as a placebo group in the following section to address this concern. Furthermore, I find no impact on vocational high-school completion rates (see Appendix Table A10).

4.3 Robustness

Results of the main regression analysis suggest that banning the veil in French schools worsens educational outcomes of Muslim women. In this section I perform a set of sensitivity analysis to validate the results, which are reported in Table 3.

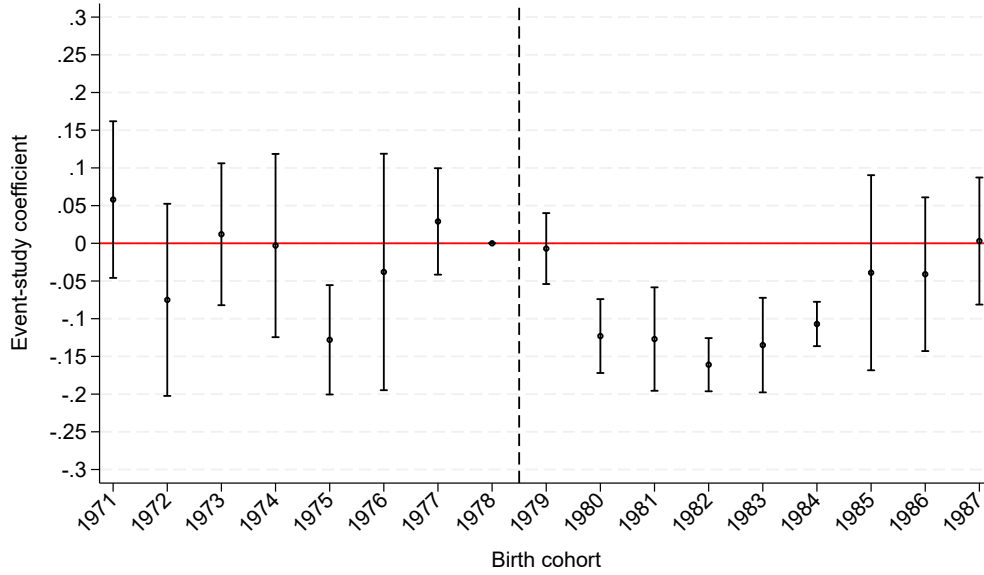
Alternative sample restrictions. First, there is empirical evidence that assimilationist policies can cause exodus of targeted groups (Saleh 2018, Saleh and Tirole 2021). We could thus be worried that Muslim pupils were more likely to study abroad as a response to the ban. In column (2), I further restrict the sample to individuals who have completed all of their education in France. The estimated impacts are only slightly lower in magnitude when I use this different sample restriction.

Second, while Muslims were the main target of this policy, it prohibited all conspicuous religious symbols. Thus, the only group that is completely unaffected by the ban is the atheists given that they have a null religiosity by definition. In the third column, I limit the control group to atheists only. As expected, the estimated drops in high-school completion rates and years of schooling are slightly larger in this specification.

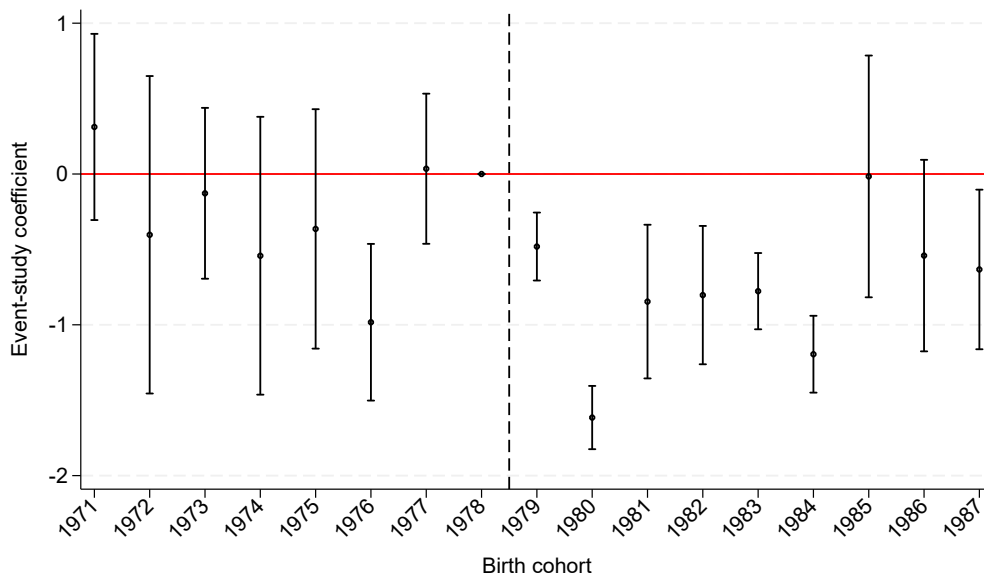
Placebo analysis. Next, in the last two columns, I estimate the difference-in-differences model using placebo groups that are unlikely to be affected by the ban. In column (4), I use the sample of men and consider Muslim men as the treatment group. As expected, since few Muslim men wear conspicuous religious symbols, I find no evidence of negative effect of the ban on high-school completion rates for this group. The apparent negative impact on years of schooling is mainly due to a decrease in vocational high-school achievement (see Appendix Table A3). Therefore, this effect could be due to the vocational high-school reform, but it could also stem from increased hostility towards Muslims in French schools more generally.

In the last column, I compare atheists to individuals reporting a non-Islamic religious affiliation. In principle, this group—mostly composed of Catholics—is potentially affected by the ban because it applies to all religious symbols. I find a precise null impact on this

Figure 2: The dynamic impact of the 1994 Bayrou circular on educational attainment of Muslim women



(a) High-school completion



(b) Years of schooling

Note: These figures plot the coefficients of event-study regressions along with 95% confidence intervals. Outcomes variables are the high-school completion rate and the number of years of post-elementary schooling (capped at 20 years). Control variables are full sets of birthyear $\times$ département of birth, survey waves, religion (or father's origins), number of siblings, and number of older siblings (capped at ten) dummies. On the right of the vertical line are cohorts who are subject to the 1994 Bayrou circular asking school principals to prohibit the headscarf in schools.

placebo group, in line with the Islamic veil being the main policy target. The inclusion of all conspicuous symbols was seen by many as “a tokenistic attempt to hide the fact that the law was really about the hijab” (Winter 2009, p. 224). Moreover, discreet symbols might be an appropriate substitute for conspicuous symbols for non-Muslims. Indeed, while most of the religious symbols worn by Muslim women are headscarves, women in other religious groups essentially only wear discreet symbols that are tolerated by law (see Table A1).

4.4 Mechanisms

The richness of the TeO surveys offers a unique opportunity to investigate the mechanisms through which the ban induces a decrease in schooling among Muslim women. Given the limited content of the Labour Force Surveys (LFS) to explore such mechanisms, Abdelgadir and Fouka (2020) turn to the first wave of the TeO surveys and a qualitative analysis of twenty in-depth interviews. Despite their small sample and the fragility of their results (as shown, for example, in Maurin and Navarrete-Hernández (2023)), their analysis highlights potential channels. They argue that the intense public debates following the ban might have spurred discrimination against Muslims. Much of the public discourse adopted an anti-veiling and anti-Muslim tone (Scott 2009, Winter 2009). In turn, increased perceptions of discrimination might have reduced school performance through a feeling of alienation (Gould and Klor 2016).

In Table 4, I explore the discrimination channel by using self-reported experiences of racism and discrimination as the outcome variable. In column (1) of Table 4, I show that treated cohorts are about 13.6 percentage points (38% of the pre-ban mean) more likely to report any experience of discrimination.

Interestingly, the TeO survey also asked respondents about the source and the context of the alleged discrimination. In columns (2) to (4), I leverage this information and I find that treated cohorts are more likely to perceive being discriminated against because of their religion. The impact on religion-based discrimination is large, at half of the pre-circular mean. Muslim affiliation being correlated with African (especially Maghrebi) origins, I unsurprisingly find a positive effect on origin-based discrimination. It is, however, much lower in magnitude as a share of the pre-circular mean (15%), suggesting that religion-based discrimination is the main factor. In column 4, I further find that these additional experiences of discrimination did occur at school. Muslim women also display higher rates of mistrust in the French school system (column 5), in line with increased hostility and a sense of alienation experienced by Muslim pupils. In Appendix Table A4, I document that Muslim men are also more likely to report being discriminated against and to mistrust the French school system. This could explain the negative impacts on years of schooling for men found in Table 3.

Table 3: Robustness checks

	Placebo groups				
	Baseline (1)	Schooling in France (2)	Muslims vs atheists (3)	Muslim men (4)	Other religions (5)
<i>Panel A: High school completion</i>					
Muslim _i × Post _c	-0.110*** (0.014) [0.066]	-0.102*** (0.016) [0.112]	-0.115** (0.008) [0.584]	0.005 (0.013) [0.910]	
Muslim _i	-0.215*** (0.035) [0.014]	-0.193*** (0.028) [0.007]	-0.150*** (0.001) [0.017]	-0.152*** (0.031) [0.037]	
Not atheist _i × Post _c					-0.000 (0.007) [0.980]
Not atheist _i					0.070* (0.034) [0.174]
Main controls	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.599	0.609	0.599	0.514	0.746
N	7,046	6,553	4,407	6,613	5,431
R ²	0.039	0.036	0.034	0.033	0.028
<i>Panel B: Years of schooling</i>					
Muslim _i × Post _c	-0.565* (0.263) [0.508]	-0.457** (0.202) [0.474]	-0.932** (0.040) [0.441]	-0.652*** (0.071) [0.044]	
Muslim _i	-1.262*** (0.352) [0.092]	-0.945** (0.397) [0.195]	-1.167*** (0.004) [0.018]	-1.330*** (0.167) [0.008]	
Not atheist _i × Post _c					-0.087 (0.059) [0.520]
Not atheist _i					0.278 (0.341) [0.555]
Main controls	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	14.646	15.046	14.646	14.373	15.651
N	6,808	6,322	4,250	6,421	5,225
R ²	0.038	0.045	0.040	0.028	0.024

Note: This Table reports results of robustness checks. In column (2) the sample is restricting to women who have studied in France. In column (3), I compare only Muslims and atheists. In columns (4) and (5), I consider placebo groups. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. Standard errors clustered at the religion × born-post-1979 level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

Another potential channel which may explain the negative impact I document could be a reaction to the ban from pupils' families. [Dahl et al. \(2022\)](#) find that, in reaction to

Table 4: Impact of the 1994 ministerial circular on experiences of discrimination

	Any experience (1)	Due to her religion (2)	Due to her origins (3)	At school (4)	Mistrust of French school (5)
Muslim _i × Post _c	0.136*** (0.019) [0.081]	0.102*** (0.021) [0.305]	0.105*** (0.029) [0.147]	0.123** (0.046) [0.262]	0.044*** (0.006) [0.047]
Muslim _i	0.084** (0.033) [0.172]	0.052 (0.107) [0.792]	0.273*** (0.080) [0.046]	-0.047 (0.066) [0.631]	-0.024 (0.025) [0.555]
Main controls	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.361	0.205	0.694	0.349	0.118
N	6,784	1,664	1,664	2,258	6,731
R ²	0.047	0.175	0.235	0.050	0.013

Note: This Table reports regression estimates of the impact of the Bayrou circular on self-reported experiences of discrimination of Muslim girls. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. Standard errors clustered at the religion level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

an integration policy in Germany, Muslim parents were less likely to help their daughters with their homework and learning, with negative consequences on girls' self-esteem and life satisfaction. In theory, a similar disinvestment in girls' education could have occurred in the French context as a reaction to the perceived secularization of education (see [Carvalho et al. 2024](#)).

To help rule out this mechanism, I first evaluate the impact of the circular on whether girls received help at school from their family and peers. In columns (1) to (4) of Table 5, I show the absence of significant impacts on whether the girl was helped by her father, her mother, her siblings, and her friends.

Second, the circular only applied to public schools and, therefore, private schools were forbidden to apply the ban. I check whether parents were more likely to send their daughters to private schools so as to avoid the veil ban. Results in column (5) suggest that this is not the case: if anything, private-school attendance may have decreased. Muslim parents were only slightly more likely to send their daughters to another school than that of their catchment area (column 6). Therefore, it is unlikely that Muslim families sending their daughters to schools which have not yet implemented the ban would drive my results. Thus, Muslim girls' disengagement from formal schooling does not appear to be driven by changes in behavior of their family members.

Third, in column (7), I evaluate whether the Bayrou circular triggered more conflict related to religion among Muslim families. I find, if anything, a reduction in such religious conflicts. Overall, these results suggest that strategic reactions from pious parents are unlikely to be driving the results.

Table 5: Impact of the 1994 ministerial circular on relatives' investments in girls' education

	Was helped at school by				Type of school		Religious conflict
	Father	Mother	Siblings	Friends	Private	Not in sector	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Muslim _i × Post _c	-0.021 (0.043) [0.828]	0.017 (0.026) [0.771]	0.038 (0.049) [0.770]	0.008 (0.021) [0.889]	-0.016** (0.006) [0.309]	0.019*** (0.006) [0.257]	-0.047*** (0.014) [0.265]
Muslim _i	-0.047 (0.091) [0.742]	-0.074 (0.137) [0.714]	-0.002 (0.034) [0.973]	-0.023 (0.036) [0.685]	-0.108*** (0.034) [0.076]	-0.085 (0.067) [0.396]	-0.057 (0.036) [0.350]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.281	0.277	0.716	0.349	0.072	0.161	0.184
N	3,325	3,371	3,228	3,423	6,784	6,776	6,778
R ²	0.048	0.132	0.293	0.035	0.191	0.017	0.012

Note: This Table reports regression estimates of the impact of the issuance of the Bayrou circular on Muslim girls' relatives' investments in their education. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. Standard errors clustered at the religion level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Heterogeneous impacts

In Table 6, I investigate sources of heterogeneity in impacts of the ban on educational attainment. The survey contains several variables on the environment in which respondents grew up. I exploit this information for this analysis. This investigation serves two purposes, namely further validating the role of parental influence and understanding how the familial context might shape long-run effects.

First, to further validate the absence of parental influence, I verify whether the estimated impact differs for women whose parents invested more in transmitting their religion. I use two measures of parental transmission of religion: whether the respondent reports religion as being “very important” in her education (at home) and name-giving.¹² For name-giving, my preferred measure is the transmission of either an Arabic-sounding first name or of a religious (Muslim) first name.¹³ Therefore, I have both a measure of *perceived* vertical transmission of religion and a measure based on revealed preferences of parents. I describe the construction of these variables in more detail in Appendix A.1.

I report results of this heterogeneity analysis in columns (1) and (2) of Table 6. I find that stronger parental religious influence, if anything, attenuates the negative effect on schooling, though this heterogeneous effect is not statistically significant at the 95% level.

¹²Name-giving has been recognized as an important cultural transmission channel ([Fryer and Levitt 2004](#), [Abramitzky et al. 2020](#), [Algan et al. 2022](#)).

¹³Including Arabic-sounding first names substantially increases power. Arabic names typically signal the Muslim religion as most of those names come from the Quran ([Algan et al. 2022](#)).

Table 6: Heterogeneous impacts on high-school completion rates

	Parental religious transmission		Parents' labor-force status		Language spoken at home	
	(1)	(2)	(3)	(4)	(5)	(6)
Muslim _i × Post _c	-0.123*** (0.010) [0.011]	-0.204*** (0.057) [0.254]	-0.114*** (0.015) [0.070]	-0.063*** (0.012) [0.169]	-0.112*** (0.012) [0.035]	-0.104*** (0.011) [0.035]
Muslim _i × Post _c × Religious education _i	0.032* (0.016) [0.422]					
Muslim _i × Post _c × Arabic first name _i		0.188* (0.095) [0.424]				
Muslim _i × Post _c × Mother was working _i			0.016** (0.006) [0.496]			
Muslim _i × Post _c × Traditional family _i [†]				-0.075*** (0.011) [0.123]		
Muslim _i × Post _c × Spoke French with mother _i					-0.004 (0.011) [0.877]	
Muslim _i × Post _c × Spoke French with father _i						-0.005 (0.007) [0.775]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.600	0.599	0.599	0.599	0.602	0.597
N	6,986	7,046	7,046	7,046	7,034	6,924
R ²	0.040	0.044	0.043	0.040	0.040	0.043

Note: This Table reports regression estimates of the heterogeneous impact of the Bayrou circular on educational attainment of Muslim women. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion dummies, a Muslim-specific linear trend, and the covariate the treatment is interacted with. Standard errors clustered at the religion level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

[†] Traditional family_i takes the value of one if the respondent's father was working and her mother was not when aged 15 years old.

This pattern holds for both measures of parental religious transmission. Thus, parental religious pressures do not appear to be the driving my results.

Third, in column (3), I interact the treatment status with dichotomous variables indicating whether the mother was working when the respondent was 15 years old. Maternal employment partly buffers the negative effect of the circular, which suggests that working mothers may have encouraged their daughters to pursue their education despite the prohibition. In column (4), I examine heterogeneity by family type. I define a dichotomous variable taking the value of one if, when aged 15 years old, the woman's mother was at home and the father was working, so as to capture traditional families in an economic sense. I find that, while the negative impact is concentrated among male-

breadwinner households, the estimated negative impact on other families only shrinks by half.

Third, another form of cultural transmission could be the language spoken at home (e.g. [Fouka 2020](#)). In columns (5) and (6), I show that the impact of the circular on high-school graduation rates is unaffected by this household characteristic. Together with the absence of impact of parental religious transmission, these results indicate that the effects of the ban are concentrated among more traditional families, but not necessarily due to vertical religious or cultural transmission. Therefore, these results might simply reflect larger impacts among pious Muslim pupils, but I cannot check this in the data since religiosity and veiling measures are only observed later in life.

4.6 Comparison to previous literature

Misclassification of treatment. In the previous sections, I document large negative impacts of the Bayrou circular on academic achievement of Muslim women. This result contrasts with previous evidence of positive effects on women of African origin in [Maurin and Navarrete-Hernández \(2023\)](#). While I show that the latter result holds in the TeO data when using their age threshold (see Appendix Table A2), the impact is of opposite sign when religion is used to identify treatment. This suggests that using origin as a proxy for religion introduces a substantial misclassification bias. While classical measurement error would only induce an attenuation bias (towards zero), measurement error in religion is necessarily non-classical. With binary measures, any measurement error must be negatively correlated with their true values ([Bound et al. 2001](#)).¹⁴

What could explain such a large misclassification bias? Observing both the misclassified and the actual treatment indicators is a crucial advantage of the TeO data. It is rare to have this information in empirical applications.

First, when using both waves of TeO, I document that individuals of African origin are more religiously mixed than the authors claim. Only 71% of women whose father is an African national are actually Muslims in my regression sample.¹⁵ One fourth of these second-generation immigrants are atheists and therefore unaffected by the Bayrou circular in my sample.

Second, recent advances in the literature on difference-in-differences with a misclassified treatment show that misclassification can cause the DID estimand to be wrongly signed ([Denteh and Kédagni 2022](#), [Negi and Negi 2025](#)). In the case of differential

¹⁴[Abdelgadir and Fouka \(2020\)](#) verify the robustness of their results to using alternative data sources, including the first wave of the Trajectories and Origins (TeO) survey in which religion is observed. However, by using only the first wave (of 2008-2009), they obtain a very small working sample of less than 2,000 observations in the regression analysis.

¹⁵This figure is even lower for Muslim women whose father is born in an African country, the proxy used in [Abdelgadir and Fouka \(2020\)](#). Less than 59% of this population is Muslim in my regression sample.

misclassification with asymmetric errors, [Denteh and Kédagni \(2022\)](#) show that the DID estimand may be wrongly signed even at relatively low rates of misclassification.¹⁶ In the case studied here, the rates of false positives (non-Muslims of African origin) are 10.4-16.6% and those of false negatives (Muslims of non-African origin) are 11.3-19.3% under the two proxies. These are comparable to the minimal rates of asymmetric errors yielding a sign reversal in [Denteh and Kédagni \(2022\)](#). Moreover, in Appendix Table A6 I find that the rates of misclassification are correlated with my main schooling outcomes. This suggests that misclassification is unlikely to be non-differential, a necessary condition to guarantee attenuation bias (i.e. bias towards zero with no sign reversal).

Third, I further test for misclassification by interacting their measure of treatment (having a father who is an African national) with a Muslim dummy. Results are reported in Appendix Table A5. While the estimates are less precise in that specification, I find that the coefficient on the origin-religion interaction is negative. This thereby suggests that using father’s nationality at birth as a proxy likely captures positive impacts on other religious groups or on atheists.

Mechanisms. Regarding the mechanisms, [Maurin and Navarrete-Hernández \(2023\)](#) state that their results are consistent with the existence of a “silent majority” of Muslim girls who do not wish to veil and are under pressure to do so by their family. This argument is often used by proponents of veil bans who argue that such policies might liberate Muslim women from oppression. However, this point is very difficult to establish with the Labour Force Surveys used in that study (see [De Giorgi 2023](#)). These data lack information about the respondents’ family or social circles.

I argue that this proposed mechanism is unlikely to hold for two reasons. First, qualitative evidence from interviews with Muslim women do not support this assertion. The vast majority of Muslim women claim that wearing the veil is their personal choice, some of them even doing so *against* their parents’ will ([Gaspard and Khosrokhavar 1995](#), [Institut Montaigne 2016](#), [Institut Français d’Opinion Publique \[IFOP\] 2019](#)). An analysis of veiling patterns in France by [Jacquet and Montpetit \(2024\)](#) further suggests that private religious motives explain a larger share of veiling behavior than the religious environment.

Second, my results suggest that parental pressures are unlikely to be a key mechanism. I find no impact of the ministerial circular on parents’ choices in Section 4.4. Moreover, the heterogeneity analysis in Section 4.5 indicates that girls raised in families that invested more in transmitting their religion do not differentially respond to the ban.

Additional replications. Last, I end this Section by briefly discussing additional replications of (i) the impact of the Bayrou circular under AF’s age threshold and sample,

¹⁶Misclassification with asymmetric errors is the case in which a misclassified dummy has different rates of false negatives and false positives. Differential misclassification refers to the case in which misclassification can be correlated with potential outcomes conditional on the true treatment.

and (ii) the impact of the 2004 law under AF’s and (iii) MN’s threshold and sample. For each replication, I consider the three measures of treatment as in Table 2. Results are reported in Appendix Tables A7 to A9. Overall, I mostly find statistically insignificant and inconsistent impacts when using these alternative specifications. I find no clear effect of the 2004 law, consistent with the prohibition having already been widely implemented following the Bayrou circular.

5 Long-term outcomes

Having established that the 1994 ministerial circular depressed schooling outcomes of Muslim women, I now explore potential effects of this policy on their social and economic integration in the long-run. I focus on two groups of outcomes, namely economic conditions and cultural integration. While it can be expected that the decreased schooling translates into worse economic outcomes in the long-run, the potential impact on social integration is a priori ambiguous. On the one hand, there could be an identity backlash to the assimilationist policy (Sakalli 2019, Fouka 2020, Carvalho et al. 2024). On the other hand, religious minorities might reduce the intensity of their religious life when facing a more secular character of public education. For example, in the case of language policy, Clots-Figueras and Masella (2013) find that the introduction of Catalan as a teaching language in Catalan schools increased the number of people who identify as Catalan and who support Catalan independence.

Cultural integration. Long-run impacts on cultural integration in Table 7 suggest that both mechanisms might be at play. In columns (1) to (3), I estimate the impact of the Bayrou circular on measures of religiosity, namely regular attendance (at least monthly) of worship events, the use of religion to self-identify, and a religiosity index (see Section 3.1 and Appendix A.1). I do not find a clear and consistent impact on religiosity later in life. While worship attendance increases, there is no discernible effect on my religiosity index. However, I document an increase in veiling rates in public spaces (column 4), which could suggest a form of religious backlash to the prohibition.

Economic theory suggests that veil bans may lead to increased segregation of pious women (Carvalho 2013). In columns (5) to (7), I further test this theoretical possibility by studying the impact of the Bayrou circular on social integration in the long-run. My first social-integration outcome (column 5) concerns friendships. I use a dichotomous variable indicating whether most of the woman’s friends have the same religious affiliation. Consistent with the positive impact on veiling and worship attendance (column 1), Muslim women exposed to the circular are more likely to form friendships with other Muslims.

Despite these indications of increased segregation, I find that affected cohorts also increase their identification with the French society. I use two measures of cultural

integration into the French society: self-reported feelings of French identity (column 6) and whether the respondent uses French as main language at home (column 7). I find positive impacts on both measures. Taken together, these results are consistent with a strengthening of both national and religious identities ([Beaman 2016](#), [Abdelgadir and Fouka 2020](#)).

Economic outcomes. Last, I evaluate long-run economic impacts in Table 8. I estimate variants of equation (1) where the dependent variable is an employment measure, a marriage-market outcome, or fertility. I find that the increase in school drop-out rates (column 7) translates into reduced employment at the time of the survey (column 1). The estimated employment impact of 10.5 percentage points is large in magnitude at more than 16% of the mean employment of unaffected cohorts. In column (2), I show that this decrease is not specific to public-sector employment for which conspicuous religious symbols are also prohibited. The negative impact on public-sector employment accounts for about 60% of the total drop, suggesting that the loss of human capital among affected cohorts translated into worse labor-market conditions in general. Moreover, while same-faith marriage rates are unchanged (column 4), the ban increases the likelihood that Muslim women have an unemployed partner (column 5). This is consistent with ample evidence of assortative mating in the marriage market (e.g. [Choo and Siow 2006](#), [Chiappori et al. 2017](#)). Finally, affected cohorts are more likely to have children (column 6), which may also drive the negative employment impacts of both parents.

Table 7: Impact of the Bayrou circular on cultural integration of Muslim women

Dep. var:	Religiosity			Cultural integration			
	Monthly worship attendance (1)	Religion to self-define (2)	Religiosity index (3)	Veiling [†] (4)	Most friends of same religion (5)	Feels French (6)	Uses French at home (7)
Muslim _{<i>i</i>} × Post _{<i>c</i>}	0.029*** (0.007) [0.104]	-0.009 (0.007) [0.797]	-0.025 (0.020) [0.683]	0.074*** (0.003) [0.079]	0.054*** (0.016) [0.273]	0.028*** (0.006) [0.353]	0.098*** (0.008) [0.184]
Muslim _{<i>i</i>}	-0.140** (0.056) [0.126]	0.048 (0.064) [0.705]	-0.038 (0.130) [0.858]	0.129*** (0.008) [0.000]	0.105 (0.087) [0.467]	-0.046*** (0.008) [0.027]	-0.062*** (0.023) [0.200]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.047	0.213	0.252	0.150	0.574	0.901	0.811
N	7,066	6,967	7,074	7,064	5,116	7,037	2,597
R ²	0.080	0.101	0.173	0.149	0.049	0.030	0.107

Note: This Table reports regression estimates of the impact of the issuance of the Bayrou circular on cultural integration of Muslim girls. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. Standard errors clustered at the religion level are reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. [†] Veiling is a dichotomous variable taking the value of 1 if the individual wears a conspicuous religious symbol (headcovering or clothing) in public spaces.

Table 8: Impact of the Bayrou circular on long-run economic outcomes of Muslim women

Dep. var:	Employment		Marriage market				
	Employed (1)	Works in public sector (2)	Lives in a couple (3)	Partner same rel. (4)	Partner works (5)	Has children (6)	School drop-out [†] (7)
Muslim _i × Post _c	-0.105*** (0.022) [0.221]	-0.060*** (0.018) [0.288]	0.081*** (0.023) [0.279]	-0.041 (0.023) [0.463]	-0.035*** (0.005) [0.125]	0.059*** (0.015) [0.240]	0.046** (0.020) [0.480]
Muslim _i	-0.134*** (0.026) [0.023]	0.010 (0.022) [0.794]	-0.008 (0.081) [0.948]	0.284** (0.125) [0.150]	-0.005 (0.011) [0.790]	0.066 (0.081) [0.603]	0.104** (0.041) [0.169]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.642	0.212	0.640	0.871	0.884	0.788	0.305
N	7,074	7,074	7,074	3,630	4,598	7,068	0.305
R ²	0.075	0.022	0.061	0.578	0.018	0.227	0.014

Note: This Table reports regression estimates of the impact of the issuance of the Bayrou circular on long-run economic integration of Muslim girls. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In column (4), the sample is restricted to women who currently have a partner. Standard errors clustered at the religion level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

[†] School drop-out is a dummy variable taking the value of one if the individual reports having stopped her schooling journey earlier than desired.

6 Conclusion

In this paper, I use unique, rich observational data on religion in France to revisit the mixed evidence on the French headscarf ban. I find that the prohibition following the issuance of a ministerial circular is associated with a large decline in the educational attainment of exposed Muslim women. Educational outcomes recover after the fifth cohort reaches veiling age under the prohibition. The impact on the number of years of schooling persists over the medium-run. I provide suggestive evidence that the negative impact of the ban operates through heightened discrimination against Muslims and increased mistrust of the French school system rather than via responses from parents. I also find suggestive evidence that the negative impacts on educational attainment translate into lower employment and impacts the marriage market in the long run. Last, I find mixed results on long-run religious and cultural integration, suggesting a strengthening of both national and religious identities.

This study also highlights that misclassifying the group targeted by an assimilationist policy might lead to incorrect conclusions. I find that the positive effects on educational attainment documented in [Maurin and Navarrete-Hernández \(2023\)](#) appear to be attributable to individuals of African origin who are not Muslim. It is therefore unlikely

that they capture any impact of the religious prohibition. Moreover, I find no impact of the 2004 ban studied in [Abdelgadir and Fouka \(2020\)](#), in line with bans having already been widely implemented after the Bayrou circular.

Overall, my results suggest that forced assimilation policies such as headscarf bans are not a successful tool to foster the economic integration of minorities and immigrants. This result is consistent with [Fouka \(2020\)](#) and [Sakalli \(2019\)](#) who similarly find that these types of policies might backfire. However, at the same time, some well-intentioned integration policies might also hamper assimilation in contexts in which minorities are strongly attached to their traditional norms ([Dahl et al. 2022](#)). Therefore, in a context of increased global migrations from countries with non-Western cultures, more work is needed to better understand which policies can durably foster their integration.

References

- Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35.
- Abdelgadir, A. and Fouka, V. (2020). Political secularism and Muslim integration in the West: Assessing the effects of the French Headscarf Ban. *American Political Science Review*, 114(3):707–723.
- Abramitzky, R., Boustan, L., and Eriksson, K. (2020). Do immigrants assimilate more slowly today than in the past? *American Economic Review: Insights*, 2(1):125–41.
- Algan, Y., Malgouyres, C., Mayer, T., and Thoenig, M. (2022). The economic incentives of cultural transmission: Spatial evidence from naming patterns across France. *The Economic Journal*, 132(642):437–470.
- Bazzi, S., Hilmy, M., and Marx, B. (2025). Religion, education, and the state. *Review of Economic Studies*, forthcoming.
- Beaman, J. (2016). As french as anyone else: Islam and the north african second generation in france. *International Migration Review*, 50(1):41–69.
- Beauchemin, C., Hamel, C., and Simon, P. (2016). *Trajectoires et origines: enquête sur la diversité des populations en France*. INED éditions.
- Beauchemin, C., Ichou, M., and Simon, P. (2023). Trajectoires et origines 2019-2020 (teo2): présentation d’une enquête sur la diversité des populations en france. *Population (édition française)*, 78(1):11–28.
- Belzil, C. and Poinas, F. (2010). Education and early career outcomes of second-generation immigrants in france. *Labour Economics*, 17(1):101–110.
- Benzer, T. (2022). Removing cultural barriers to education: State-run Islamic schools and girls’ education in Turkey. Mimeo.
- Bisin, A., Carvalho, J.-P., and Verdier, T. (2023). Cultural transmission and religion. In Sauer, R. D., editor, *The Economics of Religion*, pages 1–62. World Scientific.
- Bound, J., Brown, C., and Mathiowetz, N. (2001). Measurement error in survey data. In *Handbook of econometrics*, volume 5, pages 3705–3843. Elsevier.
- Brodeur, A., Cook, N., and Heyes, A. (2020). Methods matter: P-hacking and publication bias in causal analysis in economics. *American Economic Review*, 110(11):3634–3660.
- Brodeur, A., Lé, M., Sangnier, M., and Zylberberg, Y. (2016). Star wars: The empirics strike back. *American Economic Journal: Applied Economics*, 8(1):1–32.
- Brodeur, A., Mikola, D., Cook, N., et al. (2024). Mass reproducibility and replicability: A new hope. IZA Discussion Paper 16912.
- Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *The review of economics and statistics*, 90(3):414–427.
- Cantoni, D., Chen, Y., Yang, D. Y., Yuchtman, N., and Zhang, Y. J. (2017). Curriculum and ideology. *Journal of political economy*, 125(2):338–392.
- Carvalho, J.-P. (2013). Veiling. *The Quarterly Journal of Economics*, 128(1):337–370.
- Carvalho, J.-P., Koyama, M., and Sacks, M. (2017). Education, identity, and community: lessons from jewish emancipation. *Public Choice*, 171:119–143.
- Carvalho, J.-P., Koyama, M., and Williams, C. (2024). Resisting education. *Journal of the European Economic Association*, 22(6):2549–2597.
- Chiappori, P.-A., Salanié, B., and Weiss, Y. (2017). Partner choice, investment in children, and the marital college premium. *American Economic Review*, 107(8):2109–2167.

- Choo, E. and Siow, A. (2006). Who marries whom and why. *Journal of political Economy*, 114(1):175–201.
- Chérifi, H. (2004). Application de la loi du 15 mars 2004 sur le port des signes religieux ostensibles dans les établissements d’enseignement publics.
- Clots-Figueras, I. and Masella, P. (2013). Education, language and identity. *The Economic Journal*, 123(570):F332–F357.
- Corekcioglu, G. (2021). Unveiling the effects of a headscarf ban: Evidence from municipal jobs in turkey. *Journal of Comparative Economics*, 49(2):382–404.
- Dahl, G. B., Felfe, C., Frijters, P., and Rainer, H. (2022). Caught between cultures: Unintended consequences of improving opportunity for immigrant girls. *The Review of Economic Studies*, 89(5):2491–2528.
- De Giorgi, G. (2023). Giacomo De Giorgi discussion of: Religious symbols in schools. *Economic Policy*, 38(113):99–100.
- De La Rochebrochard, É. (1999). Les âges à la puberté des filles et des garçons en france. mesures à partir d’une enquête sur la sexualité des adolescents. *Population (french edition)*, pages 933–962.
- Denteh, A. and Kédagni, D. (2022). Misclassification in difference-in-differences models. *arXiv preprint arXiv:2207.11890*.
- Dreber, A. and Johannesson, M. (2025). A framework for evaluating reproducibility and replicability in economics. *Economic Inquiry*, 63(2):338–356.
- Fouka, V. (2020). Backlash: The unintended effects of language prohibition in U.S. schools after World War I. *The Review of Economic Studies*, 87(1):204–239.
- Fouka, V. (2022). Assimilation in historical political economy. In Jenkins, J. A. and Rubin, J., editors, *The Oxford Handbook of Historical Political Economy*, page 669–688. Oxford University Press.
- Fouka, V. (2024). State policy and immigrant integration. *Annual Review of Political Science*, 27(46).
- Fryer, R. G. and Levitt, S. D. (2004). The causes and consequences of distinctively black names. *The Quarterly Journal of Economics*, 119(3):767–805.
- Gaspard, F. and Khosrokhavar, F. (1995). *Le foulard et la République*. FeniXX.
- Gathmann, C. and Garbers, J. (2023). Citizenship and integration. *Labour Economics*, 82:102343.
- Gertler, P., Galiani, S., and Romero, M. (2018). How to make replication the norm. *Nature*, 554(7693):417–419.
- Gould, E. D. and Klor, E. F. (2016). The long-run effect of 9/11: Terrorism, backlash, and the assimilation of muslim immigrants in the west. *The Economic Journal*, 126(597):2064–2114.
- Huntington-Klein, N., Arenas, A., Beam, E., Bertoni, M., Bloem, J. R., Burli, P., Chen, N., Grieco, P., Ekpe, G., Pugatch, T., et al. (2021). The influence of hidden researcher decisions in applied microeconomics. *Economic Inquiry*, 59(3):944–960.
- Huntington-Klein, N., Portner, C. C., McCarthy, I., et al. (2025). The sources of researcher variation in economics. Working Paper 33729, National Bureau of Economic Research.
- Institut Français d’Opinion Publique [IFOP] (2019). Etude auprès de la population musulmane en France, 30 ans après l’affaire des foulards de Creil. Technical report.
- Institut Montaigne (2016). A French Islam is possible. Technical report.
- Iyer, S. (2016). The new economics of religion. *Journal of Economic Literature*, 54(2):395–441.
- Jacquet, A. and Montpetit, S. (2024). Veiling and economic integration of muslim women in france. Available at SSRN 5036386.
- Kamel, D. (2025). Between arab and white: Syrians and the u.s. naturalization law. Mimeo.
- Lleras-Muney, A. and Shertzer, A. (2015). Did the americanization movement succeed? an evaluation of the effect of english-only and compulsory schooling laws on immigrants. *American Economic Journal.Economic Policy*, 7(3):258–290.

- Lu, W. and Yurdakul, E. (2025). To veil or not to veil? assessing the removal of headscarf ban in a muslim country. *Available at SSRN 5198076*.
- MacKinnon, J. G. and Webb, M. D. (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254.
- MacKinnon, J. G. and Webb, M. D. (2018). The wild bootstrap for few (treated) clusters. *The Econometrics Journal*, 21(2):114–135.
- Mattei, P. and Aguilar, A. (2016). *Secular institutions, Islam and education policy: France and the US in comparative perspective*. Springer.
- Maurin, E. and Navarrete-Hernández, N. (2023). Behind the veil: the effect of banning the Islamic veil in schools. *Economic Policy*, 38(113):63–98.
- Miguel, E. (2021). Evidence on research transparency in economics. *Journal of Economic Perspectives*, 35(3):193–214.
- Morsy, M. (1989). *Les femmes du Prophète*. FeniXX.
- Negi, A. and Negi, D. S. (2025). Difference-in-differences with a misclassified treatment. *Journal of Applied Econometrics*, 40(4):411–423.
- Sakalli, S. O. (2019). Secularization and religious backlash: Evidence from Turkey. Working Paper.
- Saleh, M. (2018). On the road to heaven: Taxation, conversions, and the coptic-muslim socioeconomic gap in medieval egypt. *The Journal of Economic History*, 78(2):394–434.
- Saleh, M. and Tirole, J. (2021). Taxing identity: theory and evidence from early islam. *Econometrica*, 89(4):1881–1919.
- Scott, J. W. (2009). *The politics of the veil*, volume 7. Princeton University Press.
- Shofia, N. M. (2022). Why veil? religious headscarves and the public role of women. Working Paper.
- Skrondal, A. and Rabe-Hesketh, S. (2009). Prediction in multilevel generalized linear models. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 172(3):659–687.
- Squicciarini, M. P. (2020). Devotion and development: Religiosity, education, and economic progress in nineteenth-century france. *American Economic Review*, 110(11):3454–3491.
- Winter, B. (2009). *Hijab and the Republic: Uncovering the French headscarf debate*. Syracuse University Press.

A Online Appendix

A.1 Measurement of religiosity and parental religious influence

The TeO datasets contains rich information on respondents’ religious life. I first describe the variables I use to proxy for individual religiosity. I then detail how I combine those multiple measures into meaningful indices through a measurement system.

Individual religiosity. I measure individual religiosity using survey questions on the frequency of attendance of religious ceremonies, the self-reported importance of religion in the respondent’s life, whether she uses her religion to self-identify, the respect of religious dietary restrictions, and religious marriage. I list details of these variables below:

Variable name	Values	Question	Type
attendance of religious ceremonies	never; for familial ceremonies only; for religious feasts only; one or twice a month; weekly	“How often do you attend religious ceremonies?”	ordinal
importance of religion in respondent’s life	no importance; a little; quite important; very important	“What importance do you give to religion in your life today?”	ordinal
uses religion to self-identify	yes; no	“Among the following characteristics, which ones define you best? [...] Your religion?”	indicator
respect of dietary restrictions	never; sometimes; always; none (coded as a dummy if “always”)	“In your daily life, do you respect your religion’s dietary restrictions?”	indicator
religious marriage	yes; no	“Did you and your husband do a religious wedding?”	indicator

Measurement system. Since there is no natural way to combine the ordinal and indicator variables described above into meaningful indices, I formulate a measurement system. I am interested in a latent variable, *individual religiosity*, which I assume loads into the proxies listed above. I interpret those proxies as noisy measures of the associated unobserved, underlying concept. Denote by Z the vectors of proxies for individual religiosity. I assume ordinal relationships between measures $\{Z\}$ and the underlying factor $\text{IndivReligiosity}_i$ such that:

$$Z_{i,j} = \mu_{1,j} + \lambda_j \text{IndivReligiosity}_i + \varepsilon_{i,j} \quad (3)$$

where ε is a measurement error assumed to be i.i.d. and to follow an ordinal logistic distribution. As the latent factor does not have a natural scale or location, to simplify interpretations, I normalize the means of $\text{IndivReligiosity}_i$ to zero, and its variance to one. I then predict the latent factor for each individual by calculating its empirical Bayes mean ([Skron dal and Rabe-Hesketh 2009](#)).

Parental religious transmission. I measure vertical religious pressures using two variables, namely the self-reported importance of religion in the respondent’s education and name-giving behavior.

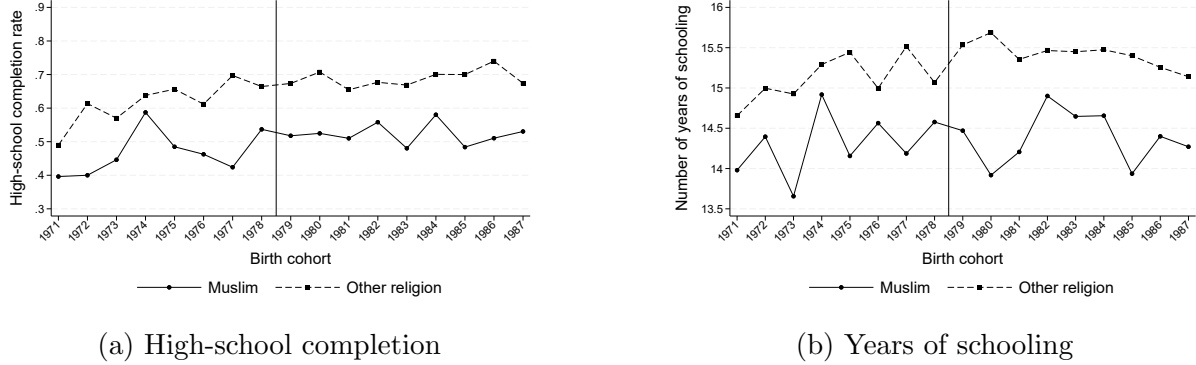
Variable name	Values	Question	Type
importance of religion in education	no importance; a little important; quite important; very important	“What importance did religion have in the education you received in your family?”	ordinal
Arabic-sounding first name	yes; no	TeO1 and constructed by the author using respondent’s first name	indicator
religious first name	yes; no	constructed by the author using respondent’s first name	indicator

The first TeO survey contains a classification of first names by likely origin. This variable is not available in TeO2, so I use all first names associated with an Arabic origin in TeO1 to impute a value to respondents in TeO2 using their first names.

I classify as religious the names of the Islamic prophet’s wives, Khadija, Sawda, Aicha, Hafsa, Zainab, Hind, Juwairiya, Safiya, Ramla, and Maimuna ([Morsy 1989](#)); and of his daughter Fatima. Variations in spelling are permitted. For male first names, I follow [Sakalli \(2019\)](#) by considering a name as religious if it is a variation of the prophet’s name (Mohamed in French) or if it begins with “Abd-” (“servant of...” in Arabic).

A.2 Appendix Tables and Figures

Figure A1: Evolution of educational attainment of Muslim and non-Muslim men across birth cohorts, 1971-1987



Note: Evolution of the high-school completion rate across birth cohorts of Muslim and non-Muslim men born in France. On the right of the vertical line are cohorts who are subject to the 1994 Bayrou circular asking school principals to prohibit the headscarf in schools.

Table A1: Summary statistics

	Muslim women			Non-Muslim women		
	Mean	SD	N	Mean	SD	N
Demographics						
Age	32.933	7.539	1,860	35.039	7.364	5,898
Born in metropolitan France	0.853	0.354	1,860	0.919	0.273	5,898
Married	0.507	0.500	1,860	0.364	0.481	5,898
Highest degree completed						
No degree	0.13	0.34	1,619	0.06	0.23	5,453
CAP/BEP [†]	0.25	0.43	1,619	0.19	0.39	5,453
High school (<i>bacc</i>)	0.28	0.45	1,619	0.22	0.41	5,453
Higher education	0.33	0.43	1,619	0.53	0.39	5,453
Economic outcomes						
Employed	0.585	0.493	1,860	0.804	0.397	5,898
Unemployed	0.176	0.381	1,860	0.096	0.295	5,898
Inactive	0.187	0.390	1,860	0.064	0.244	5,898
Years of post-elementary schooling	14.71	3.14	1,440	15.73	3.01	4,879
Religious outcomes						
Religiosity index	0.343	0.791	1,860	-0.25	0.61	5,472
Religion is very important in life	0.469	0.499	1,854	0.135	0.342	2,886
Attends religious ceremonies regularly	0.065	0.247	1,856	0.140	0.347	2,886
Had conflict over religion with parents	0.177	0.382	1,860	0.148	0.355	5,898
Most friends are of the same religion	0.700	0.458	1,860	0.805	0.396	5,898
Religion was very important in education	0.432	0.495	1,852	0.139	0.346	5,831
Partner of same religion	0.553	0.497	1,860	0.226	0.418	5,898
Wears a religious symbol	0.286	0.452	1,854	0.234	0.424	2,888
Wears a conspicuous religious symbol	0.192	0.394	1,854	<0.01	0.049	2,888

Note: The data source is the Trajectories and Origins (TeO) surveys of 2008-2009 and 2019-2020.

[†] The CAP (*Certificat d'Aptitude Professionnelle*) and the BEP (*Brevet d'Études Professionnelles*) are vocational high-school degrees aimed at acquiring skills specific to a chosen occupation (such as plumbing, butchery, or bakery).

Table A2: Impact of the 1994 ministerial circular on educational attainment using a 15-year old age threshold

Measure of treatment:	Proxies used in previous studies [†]					
	Religious affiliation (1)	(2)	Father's nationality (MN) (3)	(4)	Father's birthplace (AF) (5)	(6)
<i>Panel A: High school completion</i>						
Muslim _i × Post _c	-0.154*** (0.015) [0.032]	-0.156*** (0.012) [0.011]	0.047* (0.023) [0.192]	0.045* (0.024) [0.233]	0.073*** (0.020) [0.030]	0.072*** (0.020) [0.039]
Muslim _i	-0.232*** (0.038) [0.014]	-0.207*** (0.028) [0.005]	-0.240*** (0.033) [0.061]	-0.204*** (0.030) [0.054]	-0.082** (0.028) [0.128]	-0.077** (0.028) [0.123]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.611	0.611	0.625	0.625	0.641	0.641
N	6,905	6,905	6,818	6,818	6,872	6,872
R ²	0.039	0.076	0.046	0.080	0.043	0.078
<i>Panel B: Years of schooling</i>						
Muslim _i × Post _c	-0.812*** (0.237) [0.300]	-0.834*** (0.195) [0.212]	0.278** (0.113) [0.117]	0.261* (0.142) [0.210]	0.330** (0.102) [0.047]	0.330** (0.109) [0.058]
Muslim _i	-1.041** (0.382) [0.168]	-0.876** (0.294) [0.145]	-1.892*** (0.199) [0.020]	-1.662*** (0.196) [0.020]	-0.345* (0.185) [0.262]	-0.359* (0.194) [0.257]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	14.956	14.956	14.965	14.965	15.152	15.152
N	6,670	6,670	6,589	6,589	6,639	6,639
R ²	0.034	0.070	0.048	0.079	0.045	0.077

Note: This Table reports regression estimates of the impact of the Bayrou circular on educational attainment of Muslim girls using various measures of the treated group. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In even columns, additional control variables are indicators for whether the individual's father was working, the mother was working, and dummies for the *département* of residence at age 15. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

[†] MN refers to [Maurin and Navarrete-Hernández \(2023\)](#) and AF refers to [Abdelgadir and Fouka \(2020\)](#).

Table A3: Impact of the 1994 ministerial circular on educational attainment of men, additional controls

	High-school completion			Vocational high school		
	(1)	(2)	(3)	(4)	(5)	(6)
Muslim _i × Post _c	0.019 (0.013) [0.649]	0.017 (0.011) [0.629]	0.014 (0.010) [0.645]	-0.010 (0.010) [0.788]	-0.011 (0.008) [0.704]	-0.016* (0.008) [0.608]
Muslim _i	-0.153*** (0.030) [0.029]	-0.150*** (0.030) [0.035]	-0.129*** (0.034) [0.076]	-0.092** (0.034) [0.215]	-0.089** (0.037) [0.252]	-0.088* (0.042) [0.310]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
Parents working	No	Yes	Yes	No	Yes	Yes
Dep. at 15 y.o.	No	No	Yes	No	No	Yes
Mean dep. var.	0.516	0.516	0.516	0.769	0.769	0.769
N	6,485	6,485	6,485	6,485	6,485	6,485
R ²	0.032	0.039	0.066	0.028	0.034	0.050
<i>Panel B: Years of schooling</i>						
Muslim _i × Post _c	-0.496*** (0.076) [0.108]	-0.502*** (0.082) [0.128]	-0.493*** (0.083) [0.123]			
Muslim _i	-1.145*** (0.116) [0.003]	-1.126*** (0.122) [0.005]	-0.939*** (0.145) [0.017]			
Main controls	Yes	Yes	Yes			
Parents working	No	Yes	Yes			
Dep. at 15 y.o.	No	No	Yes			
Mean dep. var.	14.575	14.575	14.575			
N	6,296	6,296	6,296			
R ²	0.027	0.031	0.057			

Note: This Table reports regression estimates of the impact of the Bayrou circular on educational attainment of Muslim boys. Means of the dependent variable in the treatment group over pre-1979 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In column (2), additional control variables are indicators for whether the individual's father was working and the mother was working. In column (3), dummies for the *département* of residence at age 15 are also controlled for. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

Table A4: Impact of the 1994 ministerial circular on experiences of discrimination, men

	Any experience (1)	Due to his religion (2)	Due to his origins (3)	At school (4)	Mistrust of French school (5)
Muslim _i × Post _c	0.056*** (0.014) [0.308]	0.025 (0.023) [0.707]	0.123** (0.044) [0.186]	0.079* (0.037) [0.372]	0.051*** (0.008) [0.091]
Muslim _i	0.177*** (0.028) [0.012]	-0.052 (0.129) [0.793]	0.389** (0.123) [0.049]	-0.077* (0.036) [0.219]	-0.076 (0.042) [0.313]
Main controls	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	0.490	0.196	0.825	0.397	0.132
N	6,364	1,534	1,534	2,434	6,311
R ²	0.138	0.120	0.192	0.048	0.011

Note: This Table reports regression estimates of the impact of the Bayrou circular on self-reported experiences of discrimination for Muslim men. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. Standard errors clustered at the religion level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

Table A5: Heterogeneity analysis of the impact on women of African origin or with a Muslim father

Measure of treatment [†] :	Father's religion (1)	Father's nationality (MN) (2)	Father's birthplace (AF) (3)
<i>Panel A: High school completion</i>			
Muslim father _i × Post _c	-0.013 (0.036) [0.856]		
Muslim father _i × Post _c × Muslim _i	-0.056 (0.039) [0.583]		
African origin _i × Post _c		0.079*** (0.021) [0.075]	0.045* (0.023) [0.263]
African origin _i × Post _c × Muslim _i		-0.176*** (0.020) [0.014]	-0.086*** (0.025) [0.216]
Mean dep. var.	0.596	0.595	0.597
N	6,745	6,941	6,995
R ²	0.047	0.057	0.055
<i>Panel B: Years of schooling</i>			
Muslim father _i × Post _c	0.116 (0.212) [0.750]		
Muslim father _i × Post _c × Muslim _i	-0.506*** (0.071) [0.039]		
African origin _i × Post _c		0.728*** (0.124) [0.023]	0.250 (0.225) [0.481]
African origin _i × Post _c × Muslim _i		-1.228*** (0.256) [0.108]	-0.538 (0.351) [0.536]
Mean dep. var.	14.631	14.614	14.628
N	6,519	6,711	6,761
R ²	0.047	0.056	0.053

Note: Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, father's origins (nationality at birth or region of birth), and religion dummies and a Muslim-specific linear trend. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

[†] MN refers to [Maurin and Navarrete-Hernández \(2023\)](#) and AF refers to [Abdelgadir and Fouka \(2020\)](#).

Table A6: Univariate associations between misclassification and educational attainment, Muslim girls

Outcome:	High school (1)	Years of schooling (2)
$\varepsilon(\text{nationality})$	-0.057*** (0.018) [0.002]	-0.330** (0.131) [0.011]
$\varepsilon(\text{origins})$	-0.040*** (0.015) [0.009]	-0.274** (0.107) [0.010]
$\varepsilon(\text{father's religion})$	-0.109*** (0.026) [0.000]	-0.479*** (0.167) [0.004]

Note: This Table reports estimates of univariate regressions of schooling outcomes on the misclassification bias using the proxies in previous studies of the French headscarf ban or the father's religion. Following the notation in [Denteh and Kédagni \(2022\)](#), ε is an indicator taking the value of one if the treated group is misclassified under the considered proxy. The sample used is the same as in our main regression analysis. Robust standard errors in parentheses. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Replication of the impact of the Bayrou circular using AF age threshold and sample

Measure of treatment:	Proxies used in previous studies [†]					
	Religious affiliation (1)	(2)	Father's nationality (MN) (3)	(4)	Father's birthplace (AF) (5)	(6)
<i>Panel A: High school completion</i>						
Muslim _i × Post _c	0.122*** (0.013) [0.064]	0.138*** (0.010) [0.014]	0.059 (0.056) [0.527]	0.076 (0.061) [0.467]	0.033 (0.019) [0.272]	0.045* (0.024) [0.213]
Muslim _i	-0.169** (0.061) [0.160]	-0.134** (0.048) [0.130]	-0.294*** (0.030) [0.017]	-0.263*** (0.027) [0.011]	-0.116** (0.038) [0.109]	-0.125*** (0.037) [0.073]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.513	0.513	0.549	0.549	0.594	0.594
N	4,094	4,094	4,072	4,072	4,101	4,101
R ²	0.038	0.094	0.051	0.103	0.043	0.095
<i>Panel B: Years of schooling</i>						
Muslim _i × Post _c	0.283 (0.197) [0.679]	0.392* (0.210) [0.587]	0.556 (0.390) [0.351]	0.635 (0.463) [0.372]	0.437 (0.323) [0.332]	0.495 (0.312) [0.267]
Muslim _i	-0.963 (0.596) [0.365]	-0.747 (0.533) [0.397]	-1.961*** (0.147) [0.002]	-1.774*** (0.197) [0.011]	-0.408 (0.269) [0.332]	-0.482 (0.308) [0.312]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	14.365	14.365	14.463	14.463	14.880	14.880
N	3,967	3,967	3,945	3,945	3,974	3,974
R ²	0.033	0.068	0.045	0.076	0.039	0.070

Note: This Table reports replication results of the impact of the Bayrou circular on educational attainment of Muslim girls using various measures of the treated group and the empirical strategy in [Abdelgadir and Fouka \(2020\)](#). Means of the dependent variable in the treatment group over pre-1976 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In even columns, additional control variables are indicators for whether the individual's father was working, the mother was working, and dummies for the *département* of residence at age 15. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

[†] MN refers to [Maurin and Navarrete-Hernández \(2023\)](#) and AF refers to [Abdelgadir and Fouka \(2020\)](#).

Table A8: Replication of 2004 law impact using MN age threshold and sample

Measure of treatment:	Proxies used in previous studies [†]					
	Religious affiliation (1)	(2)	Father's nationality (MN) (3)	(4)	Father's birthplace (AF) (5)	(6)
<i>Panel A: High school completion</i>						
Muslim _i × Post _c	-0.013 (0.014) [0.738]	-0.021 (0.012) [0.541]	0.041 (0.034) [0.377]	0.035 (0.038) [0.478]	0.101** (0.039) [0.103]	0.093** (0.040) [0.128]
Muslim _i	-0.332*** (0.029) [0.004]	-0.305*** (0.049) [0.044]	-0.234** (0.082) [0.092]	-0.186* (0.086) [0.171]	-0.002 (0.063) [0.982]	0.028 (0.050) [0.682]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.637	0.637	0.658	0.658	0.685	0.685
N	4,920	4,920	4,845	4,845	4,895	4,895
R ²	0.031	0.067	0.033	0.067	0.031	0.066
<i>Panel B: Years of schooling</i>						
Muslim _i × Post _c	-0.028 (0.147) [0.938]	0.052 (0.088) [0.805]	0.429 (0.273) [0.259]	0.469 (0.287) [0.234]	0.721 (0.553) [0.340]	0.680 (0.548) [0.362]
Muslim _i	-1.853*** (0.290) [0.029]	-1.507*** (0.181) [0.007]	-1.750** (0.695) [0.124]	-1.314* (0.678) [0.208]	0.064 (0.755) [0.950]	0.371 (0.710) [0.695]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	15.099	15.099	15.217	15.217	15.453	15.453
N	4,651	4,651	4,581	4,581	4,627	4,627
R ²	0.071	0.116	0.081	0.121	0.076	0.118

Note: This Table reports replication results of the impact of the 2004 headscarf ban on educational attainment of Muslim girls using various measures of the treated group and the empirical strategy in [Maurin and Navarrete-Hernández \(2023\)](#). Means of the dependent variable in the treatment group over pre-1990 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In even columns, additional control variables are indicators for whether the individual's father was working, the mother was working, and dummies for the *département* of residence at age 15. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

[†] MN refers to [Maurin and Navarrete-Hernández \(2023\)](#) and AF refers to [Abdelgadir and Fouka \(2020\)](#).

Table A9: Replication of 2004 law impact using AF age threshold and sample

Measure of treatment:	Proxies used in previous studies [†]					
	Religious affiliation		Father's nationality (MN)		Father's birthplace (AF)	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: High school completion</i>						
Muslim _i × Post _c	0.042*** (0.010) [0.216]	0.033*** (0.010) [0.281]	0.020 (0.034) [0.654]	0.021 (0.034) [0.644]	0.035 (0.031) [0.390]	0.030 (0.037) [0.528]
Muslim _i	-0.329*** (0.025) [0.001]	-0.301*** (0.036) [0.004]	-0.377*** (0.038) [0.002]	-0.328*** (0.040) [0.005]	-0.152* (0.074) [0.215]	-0.136 (0.078) [0.265]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.607	0.607	0.642	0.642	0.669	0.669
N	6,097	6,097	6,023	6,023	6,079	6,079
R ²	0.038	0.080	0.046	0.084	0.044	0.083
<i>Panel B: Years of schooling</i>						
Muslim _i × Post _c	-0.277*** (0.052) [0.108]	-0.284*** (0.070) [0.185]	-0.221 (0.276) [0.547]	-0.202 (0.239) [0.528]	0.098 (0.242) [0.743]	0.119 (0.246) [0.699]
Muslim _i	-2.491*** (0.216) [0.000]	-2.268*** (0.203) [0.001]	-2.902*** (0.311) [0.001]	-2.588*** (0.288) [0.000]	-1.219** (0.418) [0.072]	-1.046** (0.422) [0.110]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	14.942	14.942	15.115	15.115	15.281	15.281
N	5,799	5,799	5,734	5,734	5,784	5,784
R ²	0.048	0.084	0.058	0.091	0.057	0.091

Note: This Table reports replication results of the impact of the 2004 headscarf ban on educational attainment of Muslim girls using various measures of the treated group and the empirical strategy in [Abdelgadir and Fouka \(2020\)](#). Means of the dependent variable in the treatment group over pre-1986 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In even columns, additional control variables are indicators for whether the individual's father was working, the mother was working, and dummies for the *département* of residence at age 15. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[†] MN refers to [Maurin and Navarrete-Hernández \(2023\)](#) and AF refers to [Abdelgadir and Fouka \(2020\)](#).

Table A10: Impact of the 1994 ministerial circular on completion of vocational high school

Measure of treatment:	Proxies used in previous studies [†]					
	Religious affiliation		Father's nationality (MN)		Father's birthplace (AF)	
	(1)	(2)	(3)	(4)	(5)	(6)
Muslim _{<i>i</i>} × Post _{<i>c</i>}	0.013 (0.014) [0.746]	0.016 (0.015) [0.701]	-0.009 (0.011) [0.564]	-0.014 (0.010) [0.361]	-0.013 (0.014) [0.462]	-0.019 (0.011) [0.213]
Muslim _{<i>i</i>}	0.048 (0.031) [0.386]	0.044 (0.030) [0.390]	-0.052*** (0.014) [0.201]	-0.054*** (0.013) [0.149]	-0.057*** (0.016) [0.066]	-0.053*** (0.015) [0.066]
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
15 y.o. controls	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.206	0.206	0.187	0.187	0.177	0.177
N	6,623	6,623	6,544	6,544	6,596	6,596
R ²	0.024	0.043	0.030	0.049	0.031	0.050

Note: This Table reports regression estimates of the impact of the Bayrou circular on completion of any high-school degree (including CAP/BEP) among Muslim girls using various measures of the treated group. Means of the dependent variable in the treatment group over pre-1980 cohorts are reported. Control variables are full sets of birthyear, survey waves, and religion (or father's origins) dummies and a Muslim-specific linear trend. In even columns, additional control variables are indicators for whether the individual's father was working, the mother was working, and dummies for the *département* of residence at age 15. Standard errors clustered at the religion (or father's origin) level reported in parentheses. *p*-values computed using the wild subcluster bootstrap procedure of [MacKinnon and Webb \(2018\)](#) are reported in brackets. Level of statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[†] MN refers to [Maurin and Navarrete-Hernández \(2023\)](#) and AF refers to [Abdelgadir and Fouka \(2020\)](#).